Jenkins Installation & Setup

Jenkins is an open-source server that is written entirely in Java. It lets you execute a series of actions to achieve the continuous integration process, that too in an automated fashion.

Jenkins facilitates continuous integration and continuous delivery in software projects by automating parts related to building, test, and deployment. This makes it easy for developers to continuously work on the betterment of the product by integrating changes to the project.

Jenkins automates the software builds in a continuous manner and lets the developers know about the errors at an early stage. A strong Jenkins community is one of the prime reasons for its popularity. Jenkins is not only extensible but also has a thriving plugin ecosystem.

Some of the possible steps that can be performed using Jenkins are:

- Software built using build systems such as Gradle, Maven, and more.
- Automation testing using test frameworks such as Nose2, PyTest, Robot, Selenium, and more.
- Execute test scripts (using Windows terminal, Linux shell, etc.
- Achieve test results and perform post-actions such as printing test reports, and more.
- Execute test scenarios against different input combinations for obtaining improved test coverage.
- Continuous Integration (CI) where the artifacts are automatically created and tested. This aids in the identification of issues in the product at an early stage of development.

At the time of what is Jenkins blog, it had close to 1500+ plugins contributed by the community. Plugins help in customizing the experience with Jenkins, along with providing support for accelerating activities related to building, deploying, and automating a project.

In this document, we will see Jenkins installation and setup for deployment on different environments.

Prerequisites

Recommended hardware configuration

- 2+ vcpus
- 8 GB+ of RAM
- 30 GB+ of drive space

Software requirements:

- Java: see the Java Requirements page
- OS: Ubuntu 20.04 LTS or higher
- Web browser: see the Web Browser Compatibility page
- For Windows operating system: Windows Support Policy

Installation of JAVA Package

$ sudo apt update

$ sudo apt install default-jdk

$ java -version

$ sudo systemctl daemon-reload (Register the Jenkins service with the command)

Installation of Jenkins Package

```
$ wget -q -O - https://pkg.jenkins.io/debian-stable/jenkins.io.key | sudo apt-key add -
$ sudo sh -c 'echo deb https://pkg.jenkins.io/debian-stable binary/ > /etc/apt/sources.list.d/jenkins.list'
$ sudo apt update
$ sudo apt install jenkins -y
```

$ sudo systemctl start jenkins (You can start the Jenkins service with the command)

$ sudo systemctl status jenkins (You can check the status of the Jenkins service using the command)

This package installation will:

- Setup Jenkins as a daemon launched on start. See `/etc/init.d/jenkins` for more details.
- Create a ‘Jenkins’ user to run this service.
- Direct console log output to the file `/var/log/jenkins/jenkins.log`. Check this file if you are troubleshooting Jenkins.
- Populate `/etc/default/jenkins` with configuration parameters for the launch, e.g `JENKINS_HOME`
- Set Jenkins to listen on port 8080. Access this port with your browser to start configuration.

Note	If your `/etc/init.d/jenkins` file fails to start Jenkins, edit the `/etc/default/jenkins` to replace the line `----HTTP_PORT=8080----` with `----HTTP_PORT=8081----` Here, "8081" was chosen but you can put another port available.

Post-installation setup wizard

Now allow 8080 port in firewall to access the Jenkins portal from the Browser.

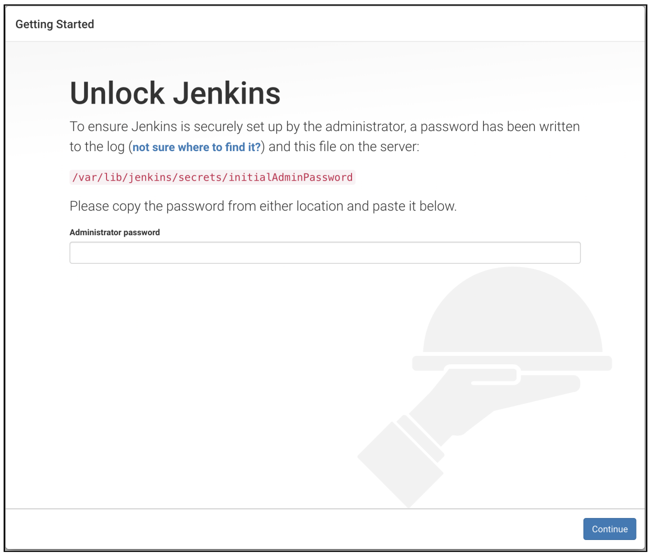
310	Port_8080	8080	Any

Now open the Browser and enter the IP address of VM with 8080 port.

For example : http://40.117.133.2:8080/

After downloading, installing, and running Jenkins, the post-installation setup wizard begins.

When you first access a new Jenkins instance, you are asked to unlock it using an automatically-generated password.



Now use the below command to get the password and paste it into above window and click on Continue.

$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword

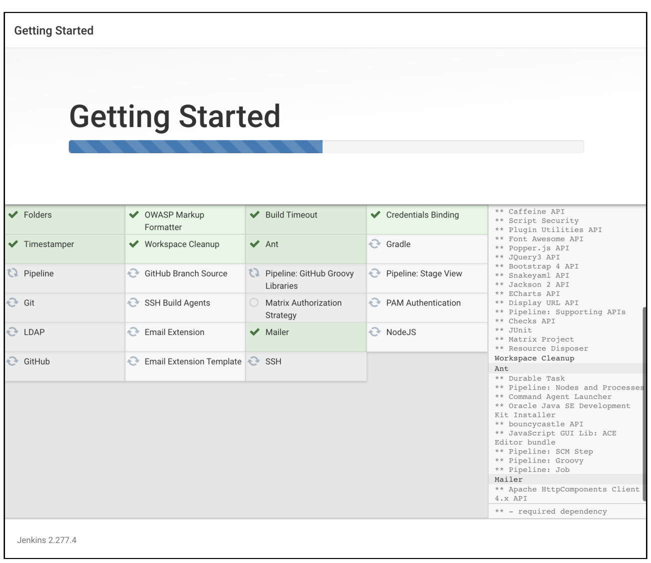
After unlocking Jenkins, the **Customize Jenkins** page appears. Here you can install any number of useful plugins as part of your initial setup.

Click one of the two options shown:

- **Install suggested plugins** - to install the recommended set of plugins, which are based on most common use cases.
- **Select plugins to install** - to choose which set of plugins to initially install. When you first access the plugin selection page, the suggested plugins are selected by default.

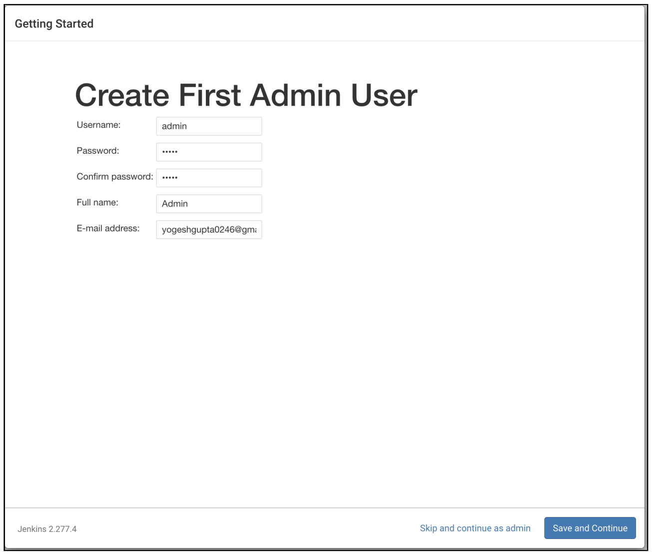
If you are not sure what plugins you need, choose **Install suggested plugins**. You can install (or remove) additional Jenkins plugins at a later point in time via the **Manage Jenkins** > **Manage Plugins** page in Jenkins.

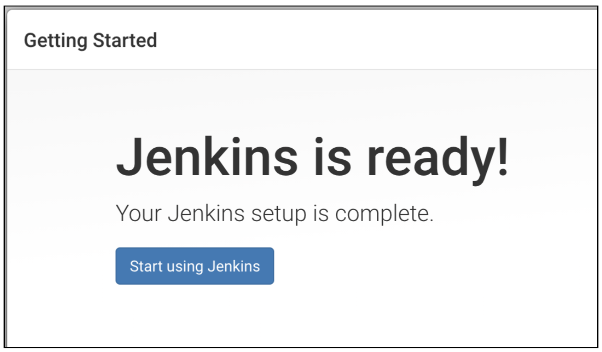
After clicking on **Install suggested plugins** you will see the below screen



Creating the first administrator user:

When the **Create First Admin User** page appears, specify the details for your administrator user in the respective fields and click **Save and Finish**.





The Jenkins setup is ready to use. Now create jobs according to your requirements.

Accessing Jenkins using URL

For now, we will be able to access Jenkins using Public IP Address and if want to access it using URL then go on respective DNS server and create an A record.

Changing default port of Jenkins

We can change the Jenkins port number from default (8080) to any custom port like 8088 etc by editing below configuration file (We need to allow this port in firewall as well)

$ sudo vim /etc/default/jenkins

If we want to access Jenkins on port 80 or 443 then we can do it in different ways. I tested in using redirection and for this, we need to create redirection entires in IP tables using below commands.

We can change the below config file as well for this:

$ sudo vim /usr/lib/systemd/system/jenkins.service

$ iptables -L -t nat

$ sudo iptables -A PREROUTING -t nat -i eth0 -p tcp --dport 80 -j REDIRECT --to-port 8080